

ABDUR RAHMAN GALIB

Full Stack Developer | AI/ML Engineer

01752817636 | abdur.galib99@gmail.com | [linkedin.com/in/abdur-rahman-07411921a](https://www.linkedin.com/in/abdur-rahman-07411921a) | github.com/galib |
portfolio-argalib9053.vercel.app
Bashundhara R/A, Dhaka

Profile Summary

Computer Science & Engineering graduate (North South University, 2026) who ships full stack web applications with real AI/ML capabilities built in, and builds independent research spanning computer vision, NLP, and LLM-based agents. Author of SAM2-MUSTAR-XAI (explainable tiny object detection), SECA (interventional credit assignment for self-evolving agents), and knowledge-augmented LLM prediction for student dropout risk, alongside shipped products like Recallo, a public AI study companion. Comfortable across Next.js/React front ends, Node/Flask/NestJS backends, and Python-based ML pipelines, end to end from research to production interface.

Technical Skills

- **Frontend & Backend:** Next.js, React.js, TypeScript, Tailwind CSS, Node.js/Express, NestJS, Flask, PHP, REST API Design, JWT/RBAC, GraphQL
- **ML/DL:** PyTorch, TensorFlow, Scikit-learn, XGBoost, CatBoost, LightGBM, CNN/Transformer architectures, AdaLoRA, Explainable AI (SHAP/LIME)
- **LLM & NLP:** LangChain, LLM orchestration (Groq, xAI, Hugging Face, Gemini, QWEN), prompt engineering, knowledge augmentation
- **Experimentation & Infra:** Bayesian optimization, Optuna, statistical testing (Fisher's exact, McNemar's), MongoDB, PostgreSQL, ChromaDB, Supabase, Docker, Linux

Featured Project

Recallo: AI-Driven Spaced Repetition & Recall Boosting Platform

Tech Stack: React, Flask, LangChain, Gemini API, XGBoost, Supabase github.com/galib/Recallo

- Built an AI study companion that uses an LLM with LangChain to auto-structure unstructured documents into topics, generate custom quizzes, and produce concept revision flashcards.
- Trained an XGBoost-based spaced repetition scheduler on a large dataset to determine optimal review intervals and trigger study session reminders.
- Implemented answer analysis that compares user responses against LLM-generated solutions to surface strong and weak areas with explanations.

AI/ML Research Projects

SHAM: Section-Aware Hierarchical Multi-Modal Malware Analysis

Tech Stack: Python, Multi-Branch CNN, Gated Fusion, XGBoost, Contrastive Learning
github.com/galib/Malware_Detection_Using_DeepLearning

- Built a multi-modal detection pipeline fusing PE section images and static header features via a gated fusion layer, followed by open-set family classification using contrastive learning and prototype embeddings.
- Added an attention-based branch-importance analysis to explain which PE section drives each family's prediction.

SECA: Safety-Constrained Evolutionary Credit Assignment

Tech Stack: Python, LLM Orchestration (Groq, xAI, Hugging Face), Statistical Testing
github.com/galib/SECA_Self_Evolving_Agent

- Identified that self-evolving LLM agents typically assign failure credit via a single unverified LLM self-report, and proposed interventional ablation, a resampled control tested against four candidate interventions, to attribute causes using statistical significance testing.
- Built a fault-injection protocol to validate attribution accuracy against ground truth, and ran a multi-cycle evolution-trajectory experiment (demo-scale, due to API budget constraints) testing whether attribution errors compound across memory cycles.

SAM2-MUSTAR-XAI: Explainable Tiny Object Detection in Aerial Images

Tech Stack: Python, PyTorch, SAM2.1 Hiera, AdaLoRA, Adaptive FPN, YOLO26

- Proposed a Mamba-enhanced, uncertainty-aware semantic-topological framework combining a SAM2.1 Hiera backbone adapted via AdaLoRA, Adaptive Feature Pyramid fusion, Gumbel-based region selection, and a dual-task uncertainty-aware ROI module for tiny object detection.
- Achieved the highest performance among evaluated architectures on the VisDrone benchmark: 70.12% precision, 57.36% F1-score, and 36.43% mAP@0.50, validated with ablation studies and XAI-based visualization of prediction-driving regions.

Enhancing Student Dropout Prediction with Knowledge-Augmented LLMs

Tech Stack: Python, XGBoost, CatBoost, QWEN, Optuna, Bayesian Optimization

- Benchmarked 11+ machine learning models (CatBoost, XGBoost, LightGBM, SVM, ensembles, and others) on a 4,424-record UCI dataset, reaching 91.32% accuracy with CatBoost after Bayesian hyperparameter optimization.
- Converted structured student records into terse key-value, prompt-based, and knowledge-augmented text formats for LLM-based prediction; a QWEN model with group-wise knowledge augmentation reached 91.19% accuracy, competitive with the best classical model.

Full Stack Projects

DaakTicket: Dynamic Blog & CMS Platform

Tech Stack: PHP, MySQL, CSS, Bootstrap github.com/galib/daakticket

- Built a full featured blogging platform with authentication, post management, comments, categories, search, an admin panel, and a responsive design across devices.

Centralized Hospital Management System

Tech Stack: React, Python, MongoDB github.com/galib/Centralized-Health-Management-System

- Built a three-portal system (doctor, patient, admin) for appointment booking and patient history tracking, implementing seven design patterns across the architecture.

Education

BSc in Computer Science & Engineering , North South University (NSU)	2022 – 2026
HSC (Science) , Ideal College	2018 – 2020